



March 14, 2025

Comment of Boston Dynamics, Inc. on the Development of an Artificial Intelligence (AI) Action Plan

Boston Dynamics, the world's leading developer of advanced mobile robots, appreciates the opportunity to comment on the development of an AI Action Plan for the United States.

The stated policy goal of Executive Order 14179 is to “to sustain and enhance America’s global AI dominance in order to promote human flourishing, economic competitiveness, and national security.” These goals cannot be achieved without sustaining dominance in **robotics**. That is because, as we explain below, robotics represents **the physical embodiment of AI**, these disciplines are rapidly converging, and the technologies will become deeply intertwined in the coming years.

Boston Dynamics, founded over 30 years ago by a Massachusetts Institute of Technology professor and his students, is at the forefront of an industry in which advanced mobile robots will become more common in both industrial and public spaces. From the outset of our research, we envisioned a future in which robots could move with the physicality and grace of people and animals and thereby become much more useful than fixed-place robots common in factories. We are best known for making the first legged robots capable of negotiating steps, stairs, rough terrain and all manner of obstacles with ease. About six years ago, we turned from decades of research towards commercialization. Today, we offer two advanced mobile manipulation robots, Spot, a quadruped robot used in factories, energy production, construction, entertainment and public safety, and a box-moving robot, Stretch, used in the logistics industry. These robots are already at work, making American industries and communities safer. For example, Spot is used by American semiconductor fabricator Global Foundries to further automate their data collection for factory condition monitoring and predictive maintenance.¹ In Cape Cod last year, a Spot deployed in a public safety mission was shot three times by an armed suspect, a result greatly preferable to having humans in harm’s way and just one of many examples of our robot keeping everyone safe during dangerous armed standoffs and hazmat operations.² Spot is also helping the US Secret Service protect the President at Mar-a-Lago.³

¹ See Global Foundries Case Study, available at <https://bostondynamics.com/case-studies/globalfoundries/>.

² See “Robotic police dog shot multiple times, credited with avoiding potential bloodshed,” Associated Press, March 27, 2024, available at <https://apnews.com/article/massachusetts-cape-cod-robot-dog-police-f63586d5286750702f396109c9a81836>.

³ See “Robotic dogs are patrolling Mar-a-Lago to help protect Trump, Secret Service confirms,” FOX 59, November 8, 2024, available at <https://fox59.com/news/national-world/robotic-dogs-are-patrolling-mar-a-lago-to-help-protect-trump-secret-service-confirms/>.

We are also working to commercialize the world’s best-known humanoid robot, Atlas, into a new commercial humanoid robot for use in manufacturing and other heavy industrial applications. Headquartered in Waltham, MA, with a workforce of approximately 850 employees, we proudly design and manufacture our robots in the United States.

As the United States government develops an Artificial Intelligence Action Plan, we have one basic but crucial comment to make: ***include robotics in that plan.***

Recent advances in AI are exciting and are already to accelerating the development of a new generation of mobile robots. Computer vision enables robots to perceive their environment, recognize objects and offer contextually relevant services. Vision as well as natural language processing will enable us to more easily communicate with robots using language and gestures. Machine learning and reinforcement learning are increasingly used to create the motion control software that drives the robots, enabling an acceleration in their development. Robots are already starting to use AI technologies to learn the tasks they should perform simply by observing and analyzing the movements of people or other robots doing that task.

Indeed, AI is already enabling our own robots perform better, in applications that matter to the US economy. Just in the past few months, we have shown the world how the AI technique of *reinforcement learning* can be used to train our Spot robot to walk with reliable stability across slippery icy or wet floors, a dramatic improvement deployed via a software update.⁴ This is development work we did in response to one of our customers, a beer manufacturing plant, whose robot frequently slipped and fell on wet factory floors. Next, we showed how visual foundation models can help Spot navigate safely around hazards in a factory setting such as ladders, electrical cords, and dolly carts.⁵ Most recently, we demonstrated how our new humanoid robot Atlas uses a combination of AI techniques, namely *reinforcement learning* and *foundation models*, to perform the complex task of part sequencing in an automotive factory setting.⁶ These are advances that are being deployed today across our robot fleet, to make American factories safer and more efficient. These are just a few examples of how artificial intelligence is converging with AI, and how any strategy relating to AI dominance must include “embodied AI,” that is to say, ***robotics***.

Of course, we aren’t the only company using AI to accelerate robot capabilities. AI and reinforcement learning are also enabling startups, including those based overseas, to catch up with robotic developments that previously took years or even decades of research and development to achieve. Like many companies based in the US who develop high-tech products, we view our greatest long-term competition as likely being Chinese tech startups. As we have

⁴ See “Starting on the Right Foot with Reinforcement Learning,” available at <https://bostondynamics.com/blog/starting-on-the-right-foot-with-reinforcement-learning/> .

⁵ See “Put It in Context with Visual Foundation Models,” available at <https://bostondynamics.com/blog/put-it-in-context-with-visual-foundation-models/> .

⁶ See “Getting Real with Humanoids,” available at <https://bostondynamics.com/blog/getting-real-with-humanoids/> .

seen in the flying robot (i.e. commercial drone) industry, over the long run and with the advantages of scaled manufacturing and favorable government policies, Chinese tech producers who are fast-followers can become market leaders.

So far, in the advanced robotics space, the Chinese have introduced products that are not nearly as good, but they are far less expensive than those developed and manufactured in the United States. In some cases, we have seen that the competition is making use of stolen intellectual property and trade secrets, with no feasible legal recourse to US-base innovators. Moreover, substantial investments are being made in China to close the gap, alongside government policies that support innovation there. While US-based robotics innovations currently lead the world, a clear threat is emerging. Another reason for concern is that products developed overseas may not be as secure as ours, and therefore are a risk in the kinds of facilities where robots such as ours are often operated, such as semiconductor fabrication plants, nuclear power plants, oil refineries, port facilities, and government buildings, or when used in agency operations. As the AI Action Plan addresses national security, the security of advanced robotics put to work in the United States ought to be a key aspect.

With the understanding that robotics is a key enabler of human flourishing, economic competitiveness, and national security, other countries have developed national robotics strategies, including China, Australia, Singapore, Japan, Korea, and Germany. In the case of China, its Ministry of Industry and Information Technology [recently announced plans](#) to dominate the humanoid robotics industry by 2027. This follows from its [five-year plan](#) announced in 2021, along with 14 other Chinese government departments, laying out a plan to position China as a global robotics leader. We have seen commensurate announcements from China of product launches,⁷ research facilities,⁸ large investments,⁹ and even plans for a humanoid robot half-marathon,¹⁰ supported by various levels of government in China.

Various organizations and experts have recently studied the growth of the Chinese robotics industry that has resulted from government support and AI-driven innovations. *See, e.g.*, “America Is Missing The New Labor Economy – Robotics,” SemiAnalysis, available at <https://semianalysis.com/2025/03/11/america-is-missing-the-new-labor-economy-robotics-part-1/> (indicating that there is “an existential threat to the US as it is outcompeted in all capacities”); “The Humanoid 100: Mapping the Humanoid Robot Value Chain,” MorganStanley, available at

⁷ How China’s government is supercharging the rise of humanoid robots,” *South China Morning Post*, February 5, 2025, available at <https://www.scmp.com/economy/china-economy/article/3297482/how-chinas-government-supercharging-rise-humanoid-robots>

⁸ China launches center to train 100-plus humanoid robots simultaneously,” *FOX News*, February 14, 2024, available at <https://www.foxnews.com/tech/china-launches-facility-train-100-plus-humanoid-robots-simultaneously>

⁹ “China’s Xpeng may invest up to \$13.8 billion in humanoid robots, state media reports,” *Reuters*, March 11, 2025, available at <https://www.reuters.com/technology/chinas-xpeng-may-invest-up-138-billion-humanoid-robots-state-media-reports-2025-03-11/>

¹⁰ “China to host world’s first half-marathon race between humans and robots,” *Sky News*, January 27, 2025, available at <https://news.sky.com/story/china-to-host-worlds-first-half-marathon-race-between-humans-and-robots-13297863>

https://advisor.morganstanley.com/john.howard/documents/field/j/jo/john-howard/The_Humanoid_100_-_Mapping_the_Humanoid_Robot_Value_Chain.pdf (“Our research suggests China continues to show the most impressive progress in humanoid robotics where startups are benefitting from established supply chains, local adoption opportunities, and strong degrees of national government support.”); “How Innovative Is China in the Robotics Industry?,” Information Technology & Innovation Foundation, available at <https://itif.org/publications/2024/03/11/how-innovative-is-china-in-the-robotics-industry/> (“China had 12 times the rate of robot use in manufacturing than did the United States.”); Testimony of Sunny Cheung, Jamestown Foundation, before the U.S.-China Economic and Security Review Commission, February 6, 2025, available at https://www.uscc.gov/sites/default/files/2025-02/Sunny_Cheung_Testimony.pdf (“The results of these sustained efforts are evident in the rapid rise of leading Chinese robotics companies such as UBTech, Fourier Intelligence, Unitree Robotics, and major tech giants like Xiaomi and XPeng, which have expanded into humanoid robotics development. These companies are increasingly competitive in global markets, not just as adopters of automation but as innovators producing cutting-edge robotic systems with intelligent AI integration.”).

The US government has not previously articulated a national robotics strategy, nor has prior federal legislation truly encouraged the use of robots and automation in key industries such as manufacturing, energy, construction, mining, and public safety. The AI Action Plan is an excellent opportunity to address these past gaps. Our recommendations for the US Government’s AI Action Plan with respect to robotics are as follows:

1. **Establish a National Strategy on Robotics.** The White House should consider issuing an Executive Order that directs the establishment of a National Strategy on Robotics. This could also be accompanied by a Robotics Commission (similar to past Executive-mandated Commissions or Committees on key technologies such as AI, biotechnology, and cybersecurity). Other prominent organizations have similarly urged the launch of a national robotics strategy. *See* Association for Advancing Automation (A3) Advocacy Principles for the United States, (“The U.S. government should develop a national strategy for robotics”), available at <https://www.automate.org/a3/advocacy-principles>; Special Competitiveness Studies Project (SCSP) Memorandum for President Trump on a National Robotics Strategy (“The White House should lead the development of a National Robotics Strategy”), available at <https://www.scsp.ai/reports/memostothepresident/robotics>.
2. **Implement Policies that Rapidly Drive Robot Deployments.** AI-empowered robotics benefit from deployment in actual work locations, collecting data that are used to train and improve those very same robots, in a method known as reinforcement learning. Policies should be pursued that drive adoption and thus kick-start this virtuous cycle of innovation, including tax incentives and support for public-private partnerships.

Government agencies themselves should be directed and funded to deploy AI-powered robotics so as to increase government efficiency.

3. **Create a Central Point of Collaboration.** In order to further develop and refine the robotics strategy, as well as to see its implementation, we envision a central point of contact in government for collaboration with industry. A White House Office of Automation and Robotics or similar central office, perhaps housed within the governing structure established to develop and execute the AI Action Plan, would serve to manage and oversee these initiatives and ensure that robotics are not overlooked as work on the AI Action Plan proceeds.

We appreciate your consideration of our perspectives and stand ready to assist the federal government in an action plan that leverages AI-powered robots to promote human flourishing, economic competitiveness, and national security.

Respectfully submitted,

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